
This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Sony
Firmware Version: N/A
Model(s): VPL-FHZ120L, VPL-FHZ90L

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.00.0000-b022	2.0.0

Version History

Module Version	Date	Notes
1_0_2_0	10/30/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly.
Example: `InterfaceName.devicePassword = 'Projector'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='VPL-FHZ120L')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='VPL-FHZ120L')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 53595, Model='VPL-FHZ120L')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value '4:3'	Value '16:9'	Value 'Full 1'
	'Full 2'	'Normal'	'Full'
	'Zoom'		
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command Focus	Value 'Far'	Value 'Near'	Value 'Focus'
# Focus example InterfaceName.Set('Focus', 'Far')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'RGB/YPbPr (BNC)'	Value 'RGB'	Value 'DVI-D'
	'HDMI'	'HDBaseT'	
# Input example InterfaceName.Set('Input', 'RGB/YPbPr (BNC)')			
Command LampMode	Value 'Standard'	Value 'Middle'	Value 'Low'
	'Extended'	'Custom'	
# LampMode example InterfaceName.Set('LampMode', 'Standard')			
Command LensLock	Value 'On'	Value 'Off'	

# LensLock example InterfaceName.Set('LensLock', 'On')			
Command LensShift	Value 'Up' 'Right'	Value 'Down' 'Shift'	Value 'Left'
# LensShift example InterfaceName.Set('LensShift', 'Up')			
Command MenuNavigation	Value 'Menu' 'Right' 'Return'	Value 'Up' 'Left'	Value 'Down' 'Enter'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command MultiScreen	Value 'On'	Value 'Off'	
# MultiScreen example InterfaceName.Set('MultiScreen', 'On')			
Command PictureMode	Value 'Dynamic' 'Multi-screen'	Value 'Standard' 'sRGB' ¹	Value 'Brightness Priority'
# PictureMode example InterfaceName.Set('PictureMode', 'Dynamic')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute	Value None		
# VideoMute example InterfaceName.Set('VideoMute', None)			
Command Zoom	Value 'Up'	Value 'Down'	Value 'Zoom'
# Zoom example InterfaceName.Set('Zoom', 'Up')			

¹Supported on model VPL-FH120L

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value '4:3' 'Full 2' 'Zoom'	Value '16:9' 'Normal'	Value 'Full 1' 'Full'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DeviceStatus	Value 'No Error' 'System Error 2' 'Lens Cover Error' 'Installation Angle Error' 'Wheel Rotation Error' 'Lens Shift Error'	Value 'Power Supply Error' 'Cover Error' 'Shock Error' 'Temperature Error' 'Luminance Error' 'Shutter Error'	Value 'Power Supply (D5V) Error' 'Light-source Error' 'Lens Not Attached Error' 'Fan Error' 'Assembling Error'
# DeviceStatus examples InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode examples			

<pre>InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)</pre>			
Command Freeze	Value 'On'	Value 'Off'	
<pre># Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)</pre>			
Command Input	Value 'RGB/YPbPr (BNC)' 'HDMI'	Value 'RGB' 'HDBaseT'	Value 'DVI-D'
<pre># Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>			
Command LampHours	Value Hours		
<pre># LampHours examples InterfaceName.Update('LampHours') Value = InterfaceName.ReadStatus('LampHours') InterfaceName.SubscribeStatus('LampHours', None, FeedbackHandler)</pre>			
Command LampMode	Value 'Standard' 'Extended'	Value 'Middle' 'Custom'	Value 'Low'
<pre># LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)</pre>			
Command LensLock	Value 'On'	Value 'Off'	
<pre># LensLock examples InterfaceName.Update('LensLock') Value = InterfaceName.ReadStatus('LensLock') InterfaceName.SubscribeStatus('LensLock', None, FeedbackHandler)</pre>			
Command MultiScreen	Value 'On'	Value 'Off'	
<pre># MultiScreen examples InterfaceName.Update('MultiScreen') Value = InterfaceName.ReadStatus('MultiScreen') InterfaceName.SubscribeStatus('MultiScreen', None, FeedbackHandler)</pre>			
Command OperationHours	Value Hours		
<pre># OperationHours examples InterfaceName.Update('OperationHours')</pre>			

Value = InterfaceName.ReadStatus('OperationHours') InterfaceName.SubscribeStatus('OperationHours', None, FeedbackHandler)			
Command PictureMode	Value 'Dynamic' 'Multi-screen'	Value 'Standard' 'sRGB' ¹	Value 'Brightness Priority'
# PictureMode examples InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)			
Command Power	Value 'On' 'Cooling 2' 'Power Saving Standby'	Value 'Startup' 'Power Saving Cooling 1' 'Off'	Value 'Cooling 1' 'Power Saving Cooling 2'
# Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			

¹Supported on model VPL-FH120L

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Ensure that the following setting in the projector menu is set for the module to work properly:

Connection → Power Menu → HDBaseT Settings → RS-232C Setting → RS-232C

Serial communication

Port Type: RS-232

Baud Rate: 38400

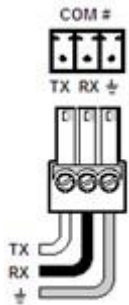
Data Bits: 8

Parity: Even

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface

Port Type:	Ethernet
Default Port:	53595
Logon Credentials Supported:	Yes
Default Password:	Projector
Multi-Connection	Undetermined
Capabilities:	
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Ensure that the follow settings in the projector are set for the module to work properly:

1. Access projector's Web Control Window by typing in the projector's IP address in a web browser and enable ADCP protocol:
Setup → Advanced Menu → ADCP → **Start ADCP Service**
2. From projector's OSD menu:
 - a. Connection → Power Menu → HDBaseT Settings → LAN Setting → **LAN Port**
 - b. Connection → Power Menu → ECO → Standby Mode → **Standard**

Appendix A. Set Commands

Aspect Ratio 16:9	aspect "16_9"\x0D\x0A
Aspect Ratio 4:3	aspect "4_3"\x0D\x0A
Aspect Ratio Full	aspect "full"\x0D\x0A
Aspect Ratio Full 1	aspect "full1"\x0D\x0A
Aspect Ratio Full 2	aspect "full2"\x0D\x0A
Aspect Ratio Normal	aspect "normal"\x0D\x0A
Aspect Ratio Zoom	aspect "zoom"\x0D\x0A
Auto Image None	apa_exec\x0D\x0A
Executive Mode Off	controlkey_lock "off"\x0D\x0A
Executive Mode On	controlkey_lock "on"\x0D\x0A
Focus Far	key "lens_focus_far"\x0D\x0A
Focus Focus	key "lens_focus"\x0D\x0A
Focus Near	key "lens_focus_near"\x0D\x0A
Freeze Off	freeze "off"\x0D\x0A
Freeze On	freeze "on"\x0D\x0A
Input DVI-D	input "dvi1"\x0D\x0A
Input HDBaseT	input "hdbaset1"\x0D\x0A
Input HDMI	input "hdm1"\x0D\x0A
Input RGB	input "rgb2"\x0D\x0A
Input RGB/YPbPr (BNC)	input "rgb1"\x0D\x0A
Lamp Mode Custom	light_output_mode "custom"\x0D\x0A
Lamp Mode Extended	light_output_mode "extended"\x0D\x0A
Lamp Mode Low	light_output_mode "low"\x0D\x0A
Lamp Mode Middle	light_output_mode "mid"\x0D\x0A
Lamp Mode Standard	light_output_mode "high"\x0D\x0A
Lens Lock Off	lens_lock "on"\x0D\x0A
Lens Lock On	lens_lock "off"\x0D\x0A
Lens Shift Down	key "lens_shift_down"\x0D\x0A
Lens Shift Left	key "lens_shift_left"\x0D\x0A
Lens Shift Right	key "lens_shift_right"\x0D\x0A
Lens Shift Shift	key "lens_shift"\x0D\x0A
Lens Shift Up	key "lens_shift_up"\x0D\x0A
Menu Navigation Down	key "down"\x0D\x0A
Menu Navigation Enter	key "enter"\x0D\x0A
Menu Navigation Left	key "left"\x0D\x0A
Menu Navigation Menu	key "menu"\x0D\x0A

Global Scripter Module Communication Sheet

Menu Navigation Return	key "return"\x0D\x0A
Menu Navigation Right	key "right"\x0D\x0A
Menu Navigation Up	key "up"\x0D\x0A
Multi Screen Off	multi_screen "off"\x0D\x0A
Multi Screen On	multi_screen "on"\x0D\x0A
Picture Mode Brightness Priority	picture_mode "brt_priority"\x0D\x0A
Picture Mode Dynamic	picture_mode "dynamic"\x0D\x0A
Picture Mode Multi-screen	picture_mode "multi_screen"\x0D\x0A
Picture Mode sRGB	picture_mode "srgb"\x0D\x0A
Picture Mode Standard	picture_mode "standard"\x0D\x0A
Power Off	power "off"\x0D\x0A
Power On	power "on"\x0D\x0A
Video Mute None	key "blank"\x0D\x0A
Zoom Down	key "lens_zoom_down"\x0D\x0A
Zoom Up	key "lens_zoom_up"\x0D\x0A
Zoom Zoom	key "lens_zoom"\x0D\x0A

Appendix B. Update Commands

Aspect Ratio	aspect ?\x0D\x0A
Device Status	error ?\x0D\x0A
Executive Mode	controlkey_lock ?\x0D\x0A
Freeze	freeze ?\x0D\x0A
Input	input ?\x0D\x0A
Lamp Hours	timer ?\x0D\x0A
Lamp Mode	light_output_mode ?\x0D\x0A
Lens Lock	lens_lock ?\x0D\x0A
Multi Screen	multi_screen ?\x0D\x0A
Operation Hours	timer ?\x0D\x0A
Picture Mode	picture_mode ?\x0D\x0A
Power	power_status ?\x0D\x0A